HIGHER SECONDARY SECOND YEAR(+2)

PUBLIC EXAMINATION – MARCH 2025

COMPUTER Science – KEY ANSWERS

PART-I

Answer All the Questions **15 x 1=15**

Q. NO	OPTION	ANSWER	Mark
1	c	Definition	1
2	a	Constructors	1
3	b	LEGB	1
4	c	Half interval search	1
5	b	Ternary	1
6	a	For	1
7	d	return	1
8	a	Positive or negative	1
9	a	ecneicS retupmoC	1
10	d	.	1
11	b	Chen	1
12	b	Select	1
13	a	modification	1
14	b	Os module	1
15	b	distinct	1

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PART-II

Answer any six questions. Question no 24 is compulsory **6x2=12**

Q. NO	ANSWER	MARKS
16.	Bundling two values together into one can be considered as a pair **(or)** Pair is a compound structure which is made up of list or Tuple **(or)** Bundling two values together into one can be considered as a pair. Any suitable example	2
17.	Sorting is a process of arranging group of items in ascending or descending order Types: Bubble sort, selection sort, insertion sort	2

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18.	5 10 15	**2**
19.	Slice is a substring of a main string. A substring can be taken from the original string by using [ ] operator and index or subscript values	**1** **1**
20.	Destructor is also a special method to destroy the objects. In Python, _ _del_ _( ) method is used as destructor. (or) def_del_(self): <statements>	**2**
21.	DDL Command: Create-To create tables in a database DML command: Insert-Inserts data into table	**2**
22.	Cd command refers to change directory and absolute path refers to the complete path where python is installed.	**2**
23.	Pip is a Package manager software for installing python packages. Importing Matplotlib using the command: import matplotlib. pyplot as plt Matplotlib in your workspace	**2**
24.	False Welcome	**2**

PART-III

Answer any six questions. Question no 33 is compulsory 6x3=18

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	ANSWER	MARKS
25.	The variables used inside the function may cause side effects though the functions which are not passed with any arguments In such cases the function is called impure function. When a function depends on variables or functions outside of its definition block, you can never be sure that the function will behave the same every time it's called.	**3**
26.	A variable which is declared outside of all the functions in a program is known as global variable. This means, global variable can be accessed inside or outside of all the functions in a program. **(or)** A variable, with global scope can be used anywhere in the program. It can be created by defining a variable outside the scope of any function Any suitable example	**2** **1**
27.	1. = is a simple assignment operator to assign values to variable 2. There are various compound operators in Python like +=, -=, *=, /=, %=, **= and //= are also available Any suitable example	**2** **1**

28. (3)

ceil ()	floor ()
Returns the smallest integer greater than or equal to x	Returns the largest integer less than or equal to x
Syntax: math.ceil (x)	Syntax: math.floor (x)
Example: print(math.ceil(26.7)->27 **(or) A suitable Example**	Example: print(math.floor(26.7)->27 **(or) A suitable Example**

29. **Syntax:** **3**

(i)List.remove(element)
(ii)List.pop(index of an element)
(iii) List.clear( )

30. Cross product is a way of combining two relations. The resulting relation contains, both relations being combined. **2**

A x B means A times B, where the relation A and B have different attributes.

Suitable example **1**

31. **3**

reader()	DictReader( )
Reads CSV files into a list	Read CSV files into a dictionary
csv.reader works with list/tuple.	csv.DictReader works with dictionary
Takes each line of the file and make a list of all columns	Takes additional argument fieldnames that are used as dictionary keys

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32. fetchone (): The fetchone () method returns the next row of a query result set or None in case there is no row left. 3
fetchmany () method that returns the next number of rows (n) of the result set

33. [1,4,9,16,25] **3**

PART-IV

Answer All the following questions **5x5=25**

Q. NO	ANSWER	MARKS
34. (a)	Pure function *Pure functions are functions which will give exact result when the same ar* A function can be a pure function provided it should not have any external variable which will alter the behaviour of that variable *guments are passed* *Example* *Impure functions* The variables used inside the function may cause side effects though the functions which are not passed with any arguments	$2^{1/2}$ $2^{1/2}$

	When a function depends on variables or functions outside of its definition block, you can never be sure that the function will behave the same every time it's called Example	
(b)	Hierarchical Model Relational Model Network Database Model Entity Relationship Model Object Model **Explanation of each model**	**1** **4**
35.(a)	Operators are categorized as 1. Arithmetic operators 2. Relational operators (or) Comparative operators 3. Logical operators 4. Assignment operators 5. Conditional operator(or) Ternary operator Explanation of each operator with suitable example	**1** **4**
(b)	1.Unique constraint 2.Primary Key constraint 3.Default constraint 4.Check constraint Explanation of each constraint	**1** **4**
36.(a)	Syntax: *while <condition>:* *statements block 1* *[else:* *statements block2]* Any suitable example &Explanation	**2** **3**
(b)	'Explain any 5 modes with explanation r','w','a','x','t','b','+'	**5**
37.(a)	When a function calls itself is known as recursion. Recursion works like loop but sometimes it makes more sense to use recursion than loop. You can convert any loop to recursion.	

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	A recursive function calls itself. Imagine a process would iterate indefinitely if not stopped by some condition! Such a process is known as infinite iteration. The condition that is applied in any recursive function is known as base condition. A base condition is must in every recursive function otherwise it will continue to execute like an infinite loop. Any suitable Example	**5**
(b)	getopt,sys,os Syntax Definition Example	**5**
38.(a)	{1,2,3,4,5,6,7,8} {4,5} {1,2,3} {6,7,8} {1,2,3,6,7,8}	**5 Each 1 mark**
(b)	(i)AVG () The SQL AVG () function returns the average of row/column data in a table (ii) COUNT () The SQL COUNT () function returns the number of rows in a table satisfying the criteria specified in where clause (iii)SUM () The SQL SUM () function returns the sum of row/column data in a table (iv) MAX () The MAX () function returns the largest value of the selected column. (v) MIN () The MIN() function returns the smallest value of the selected column Any suitable example	**5 Each 1 mark**

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